

# Erik Olvera

Software Engineer | Python, FastAPI & Full-Stack Applications

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## EDUCATION

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**Texas State University** | B.S. in Computer Science | May 2026

Cum Laude | GPA: 3.51 | Minor: Data Analytics

BOBCATshadow: Dell Technologies and Charles Schwab; exposure to enterprise SDLC and Agile.

## TECHNICAL SKILLS

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**Languages:** Python, TypeScript, JavaScript, SQL

**Backend & data:** FastAPI, Pydantic, REST APIs, OpenAPI, PostgreSQL, Supabase, Redis

**Applications & AI:** Next.js, React, Tailwind CSS, CrewAI, Gemini API

**Testing & delivery:** pytest, Hypothesis, Vitest, Git, GitHub Actions, Docker, Vercel

## SELECTED PROJECTS

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### DebtPilot | Debt payoff planner

Python, FastAPI, Pydantic, Next.js, TypeScript | [Live application](#) | [Source code](#)

- Built a pure Python simulator shared by Snowball, Avalanche, and minimum-only strategies. Preserved cent precision with Decimal, payment rollover rules, and bounded simulations.
- Normalized income frequencies and capped extra payments to available cash. Enforced string-valued money at the API boundary and generated TypeScript types from OpenAPI.
- Validated financial invariants, hand-calculated schedules, and an independent amortization oracle. Local audit: 260 backend tests passed; 98.72% coverage with branch measurement enabled (Sep. 7, 2026).

### Gaffer AI (GOLAZO) | AI match briefing service

Python, FastAPI, CrewAI, Gemini API, Pydantic, Redis | [Source code](#)

- Extended a CrewAI scaffold into a FastAPI service with sequential research, analysis, and writing agents. Used typed models to validate research data and final response structure.
- Added memory and optional Redis caching, a second cache check inside a process lock, and scheduled fixture pre-generation to reduce duplicate, quota-limited AI calls.
- Implemented model fallback chains, retry sweeps, and graceful Redis-to-memory fallback; configured a Render deployment with pinned dependencies.

### AniMood | Anime discovery application

Four-person team | Next.js, React, Supabase, Gemini API, Jikan API | [Live application](#)

- Built MoodBot's chat interface and AI integration, translating mood input into structured genre parameters for Jikan queries, with sequel filtering and error fallbacks.
- Implemented global search and watchlist features; established signup validation and early automated tests. Added MoodBot component and AI-service tests with mocked network calls.
- Migrated the Vite application to Next.js with server-rendered detail pages, cookie-based Supabase sessions, route protection, and a server-side Gemini API handler.

## EXPERIENCE

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### Target | Interim Team Lead & Fulfillment Pace Setter

Cypress, TX | Aug. 2022 - Present

- Served as Interim Team Lead for over a year, coordinating a fulfillment team of 15+ employees and setting daily performance standards.
- Managed resource allocation, device tracking, and team communication in leadership's absence to address workflow bottlenecks.